



Hifz Quran

Project Team

NAME	ROLL-NO
ABDUL SUBHAN	21I-1223
SAMI IRSHAD	21I-1103
MUHAMMAD SHAMMAZ	21I-1752

Session 2021-2025

Supervised by

Mr. Muhammad Amir Gulzar

Department of Computer Science

National University of Computer and Emerging Sciences

Islamabad, Pakistan

June, 2025

Contents

Project Team	1
Supervised by	1
CHAPTER 1: Introduction	1
1.1 Problem statement	1
1.2 Scope	1
1.3 Modules	3
Module 1: User Profile and Progress Management	3
Module 2: Ayah Segmentation and Memorization	3
Module 3: Personalized Learning Path	3
Module 4: Interactive Memorization Modes	3
Module 5: Mutashabihat Detection and Management	4
Module 6: Real-Time Recitation Feedback	4
Module 7: Notifications and Reminders	4
1.4 Table 1.1: User classes and characteristics	5
CHAPTER 2: Project requirements	6
2.1 Use-case/event response table/storyboarding	6
Table 1.2 User Registration and Login	6
Table 1.3: Profile Customization	7
Table 1.4: Set and Track Memorization Goals	8
Table 1.5: Ayah Segmentation for Memorization	9
Table 1.6: Detect and Manage Mutashabihat	10
Table 1.7: Adaptive Learning Path Creation	11
Table 1.8: Interactive Memorization Modes	12
Table 1.9: Receive Real-Time Recitation Feedback	13
Table 1.10: Receive Notifications and Reminders	14
2.2 Functional requirements	15
Module 2.2.1: User Profile and Progress Management	16
Module 2.2.2: Personalized Learning Path	16
Module 2.2.3: Ayah Segmentation and Memorization	16
Module 2.2.4: Interactive Memorization Modes	16
Module 2.2.5: Real-Time Recitation Feedback	17
Module 2.2.6: Mutashabihat Detection and Management	17
Module 2.2.7: Notifications and Reminders	17
2.3 Non-functional requirements	18
Performance	18
Usability	18
Reliability	18
Security	18
2.4 Figure 2.1: Domain model	19
CHAPTER 3: System overview	9
3.1 Architectural design	9

3.1.0 Architectural Design for Quran Memorization Assistant System	9
3.1.1 Modules Overview	9
3.1.2 Module Relationships and Interaction	10
3.1.3 MVC Architecture	11
3.1.4 High-Level Box and Line Diagram	11
1. View Layer (User Interface):	11
2. Controller Layer:	12
3. Model Layer:	12
3.1.5 Conclusion	13
3.2 Design models	14
Figure 3.2: Activity Diagram	14
Figure 3.3: Data Flow Diagram	15
Figure 3.4: System Sequence Diagram	16
3.3 Data design	17
3.3.1 Introduction	17
3.3.2 Data Storage Technologies	17
3.3.3 Data Entities and Structures	17
3.3.4 Data Relationships	20
3.3.5 Data Processing and Workflow	21
3.3.6 Security and Data Integrity	22
3.3.7 Figure 3.5 Data Flow Diagram	23
3.3.8 Conclusion	23

List of Figure:

1. Chapter 1
 - 1.1. User modules
2. Chapter 2
 - 2.1. Domain model
3. Chapter 3
 - 3.1. MVC
 - 3.2. Activity Diagram
 - 3.3. Data Flow Diagram
 - 3.4. System Sequence Diagram
 - 3.5. Data Flow Diagram

List of Tables:

- 1.1. USER CLASSES AND CHARACTERISTICS
- 1.2. User Registration and Login
- 1.3. Profile Customization
- 1.4. Set and Track Memorization Goals
- 1.5. Ayah Segmentation for Memorization
- 1.6. Detect and Manage Mutashabihat
- 1.7. Adaptive Learning Path Creation
- 1.8. Interactive Memorization Modes
- 1.9. Receive Real-Time Recitation Feedback
- 1.10. Receive Notifications and Reminders

CHAPTER 1: Introduction

The product described in the document is an intelligent Quran memorization assistant aimed at helping users memorize the Quran more effectively. The application is designed to adapt to individual learning preferences, pace, and achievement levels, providing an engaging and personalized experience for learners of all ages. By offering a blend of traditional memorization techniques with modern, technology-assisted tools, the system aims to make the memorization process more efficient, especially for those who may not have access to instructors or need to study independently.

Target readers of the document:

System users(memorizers), project supervisor, project team, content advisor, FYP project panel

1.1 Problem statement

Memorization of the Quran involves using techniques that do not take into consideration the styles that every student would prefer when memorizing, which make it hard for them to memorize. It is challenging for learners to be consistent; it becomes hard for them to identify and memorize similar verses known as Mutashabehat, and most importantly, they do not get any feedback on the accuracy of their recitation. There is little in the way of a resource that offers the tried-and-tested approach along with powerful learning assistances for consistent memorization. This lack of a targeted approach for learner support can hamper these goals and keep students from achieving their objectives in the shortest amount of time possible

1.2 Scope

The Project Intelligent Quran Memorization Assistant is focused on building a mobile application for people. The Quran data will be divided into small portions that will be easier to learn by heart to meet the goal set by the Quran memorization program. With the technology, users will again and again revisit the verses, and the significance of each ayah will be emphasized through individual memorization. The application will also have adaptable features with respect to learning styles and rates to ensure that the system develops learning paths for memorization. The application will also include a detecting

mechanism that will help in searching and flagging Mutashabehat like ayahs. This will help the users to differentiate between similar verses, which is a general issue whenever one tries to memorize the Holy Quran. The features that distinguish its use from other forms of learning include the different modes of recitation, listening, and, at times, both. Doing so, the ways of interaction with the Quran are also various, which contributes to the learning process with all types of activities in mind. The system will also give the user real-time feedback on his or her recitation and memorization attempts. In this way, it will provide constructive feedback and suggestions the users need to make it more accurate and fluent in the memorizing process.

The app is to be set in such a manner that it should be able to recognize the learning style and speed of each learner with the goal of enabling the creation of unique memorization routes. It will include self-adjusting repetitive strategies, thus giving the learner repetitive practice depending on their performance as to enhance memory. Additional feedback will be given in real-time through the integrated speech-to-text facility to allow the user to correct mistakes as she or he recites. Also, the platform will include the Memorize and Review modes, as well as the Recitation and Listening modes to address learners' preferences in revision. One noteworthy functionality of the application would be being able to identify Mutashabihat or similar verses that often appear to be ambiguous, thus helping the users pay more attention to these particular areas. Through the integration of these functionalities in this project, it is envisioned to provide a new horizon towards the advanced educational technology in Islamic learning where Quran memorization becomes more fun, efficient, and easily achievable

1.3 Modules

Module 1: User Profile and Progress Management

Through this module, users can set and achieve personal goals, create and maintain their profiles, and monitor their courses of Quran memorization.

- **Feature 1:** User registration and login.
- **Feature 2:** Profile customization (name, profile picture, preferred learning style)
- **Feature 3:** Progress tracking (memorized verses, review history).
- **Feature 4:** Goal setting and reminders.
- **Feature 5:** Personalized dashboard with performance analytics.

Module 2: Ayah Segmentation and Memorization

This module helps with memorization by breaking verses down into smaller, easier-to-manage chunks.

- **Feature 1:** Automatic segmentation of Ayahs into phrases
- **Feature 2:** Guided memorization of segmented phrases
- **Feature 3:** Option to merge or split segments based on user preference
- **Feature 4:** Progress tracking for each segment

Module 3: Personalized Learning Path

This module caters for the learning ability and the speed of the user by making the memorization path unique to the individual.

- **Feature 1:** Learning style analysis (listening vs reciting)
- **Feature 2:** Adaptive repetition strategies based on performance
- **Feature 3:** Customized memorization plan generation

Module 4: Interactive Memorization Modes

This module provides multiple modes to help users participate in the process of memorization.

- **Feature 1:** Memorization mode with guided steps
- **Feature 3:** Recitation mode for real-time practice
- **Feature 4:** Listening mode with playback and repeat functions

Module 5: Mutashabihat Detection and Management

This module finds and emphasizes verses that are similar and could be confusing.

- **Feature 1:** Detection of mutashabihat (similar verses)
- **Feature 2:** Special focus mode for practicing similar verses
- **Feature 3:** User guidance on differentiating similar verses
- **Feature 4:** Customized repetition strategies for similar verses

Module 6: Real-Time Recitation Feedback

Using speech recognition technology, this module gives the user immediate feedback on how well they are reciting sentences.

- **Feature 1:** Speech recognition for real-time error detection
- **Feature 2:** Immediate feedback on memorization quality of the user
- **Feature 3:** Highlighting errors and suggesting corrections (words are correctly recited)
- **Feature 4:** Performance analytics based on recitation accuracy

Module 7: Notifications and Reminders

This module ensures users stay on track with their memorization goals through timely alerts and reminders.

- **Feature 1:** Daily, weekly, and monthly reminders
- **Feature 2:** Progress-based notifications
- **Feature 3:** Goal achievement alerts

1.4 Table 1.1: User classes and characteristics

User class	Description
User	This class represents a user of the application who is involved in the Quran memorization process. It manages user-related data, such as personal information, goals, and progress
Ayah Segmentation	This class handles the division of Quranic verses (Ayahs) into smaller, manageable segments to facilitate easier memorization
Progress Tracking	This class is responsible for tracking the user's memorization progress, including verses memorized and review history
Mutashabihat	This class deals with detecting and managing mutashabihat , which are verses in the Quran that are very similar and may be confusing during memorization
Memorization Mode	This class defines various modes through which the user can engage with Quran memorization, such as guided memorization, recitation, and listening
Feedback	This class provides real-time feedback on the user's recitation using speech recognition technology, highlighting mistakes and offering suggestions for improvement
Learning style	This class determines and adapts to the user's learning style, ensuring that the memorization plan aligns with their strengths
Notification/Reminder	This class manages notifications and reminders to help users stay on track with their memorization goals and review schedules

CHAPTER 2: Project requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/event response table/storyboarding

The selection of the requirement gathering technique(s) will depend on the type of project. For instance, • Use case (use case diagram + detail use case) is an effective technique for interactive end-user applications. • Event- response table is for a real-time system in which most of the functionalities are performed at backend. • Storyboarding for graphically intensive applications.

Create the suitable representation here.

Table 1.2 User Registration and Login

use case id	UC-01
use case name	User Registration and Login
Actor	System user
description	A new user can create an account, or an existing user can log in to access their Quran memorization profile and progress
Flow	1. User provides registration details (username, email, password). 2. System validates the information. 3. If valid, the system registers the user and allows them to log in. 4. User log in with their credentials. 5. System authenticates the user and grants access to their personalized dashboard

pre condition	The user must provide valid credentials (username and password).
post condition	User is logged in and can access the system

Table 1.3: Profile Customization

use case id	UC-02
use case name	Profile Customization
Actor	System user
description	Users can customize their profile by updating their name, profile picture, and preferred learning style
Flow	1. User navigates to the profile settings page. 2. User update profile details (name, profile picture, learning style). 3. System validates the inputs and updates the profile.
pre condition	The user must be logged in.
post condition	User's profile is updated successfully.

Table 1.4: Set and Track Memorization Goals

use case id	UC-03
use case name	Set and Track Memorization Goals
Actor	System user
description	Users can set personal memorization goals (e.g., memorizing a specific number of verses) and track their progress
Flow	1. User sets a memorization goal (e.g., number of Ayahs or Surahs). 2. System records the goal and sets reminders or notifications. 3. As the user memorizes verses, the system updates progress. 4. The user can view the goal completion percentage on their dashboard.
pre condition	The user must be logged in.
post condition	Goal progress is tracked and updated in real time

Table 1.5: Ayah Segmentation for Memorization

use case id	UC-04
use case name	Ayah Segmentation for Memorization
Actor	System user
description	The user must select an Ayah for memorization
Flow	1. User selects an Ayah from the Quran. 2. System automatically segments the Ayah into smaller parts. 3. User memorizes the segments with guidance from the system.
pre condition	The user must select an Ayah for memorization
post condition	Ayah is segmented and displayed in smaller parts for easier memorization

Table 1.6: Detect and Manage Mutashabihat

use case id	UC-05
use case name	Detect and Manage Mutashabihat (Similar Verses)
Actor	System user
description	The system detects similar verses (mutashabihat) and helps the user practice them to avoid confusion.
Flow	1. User encounters or selects similar verses. 2. System detects the mutashabihat and highlights them. 3. System provides special guidance on differentiating these verses
pre condition	The user is memorizing or reviewing similar verses
post condition	User receives assistance in memorizing and differentiating similar verses

Table 1.7: Adaptive Learning Path Creation

use case id	UC-06
use case name	Adaptive Learning Path Creation
Actor	System user
description	The system adapts the learning path to the user's learning style and performance
Flow	1. System analyzes user behavior to detect their learning style (listening vs. reciting). 2. System generates an adaptive memorization path based on the user's performance and preferences. 3. User follows the customized plan, with suggestions for optimal learning methods
pre condition	The user must have completed some memorization tasks for the system to analyze their learning style.
post condition	A personalized learning path is created and followed by the user.

Table 1.8: Interactive Memorization Modes

use case id	UC-07
use case name	Interactive Memorization Modes
Actor	System user
description	Users can engage in different memorization modes such as guided memorization, recitation, or listening
Flow	1. User selects a memorization mode (e.g. recitation, listening). 2. System provides a step-by-step process for the selected mode. 3. User follows the steps, practicing either by reciting or listening.
pre condition	The user must be logged in and have access to a specific Ayah or Surah
post condition	User completes the memorization practice in the selected mode

Table 1.9: Receive Real-Time Recitation Feedback

use case id	UC-08
use case name	Receive Real-Time Recitation Feedback
Actor	System user
description	The system provides real-time feedback to the user's recitation using speech recognition.
Flow	1. User recites an Ayah or verse. 2. System listens to the recitation using speech recognition technology. 3. System compares the recitation to the correct version of the verse. 4. System provides feedback, highlighting errors and suggesting corrections
pre condition	The user is reciting an Ayah
post condition	User receives real-time feedback on their recitation accuracy

Table 1.10: Receive Notifications and Reminders

use case id	UC-09
use case name	Receive Notifications and Reminders
Actor	System user
description	Users receive reminders and notifications to help them stay on track with their memorization schedule
Flow	1. User sets up daily, weekly, or goal-based reminders. 2. System sends notifications when it is time to review or memorize. 3. System sends notifications based on progress or goal achievements.
pre condition	The user must have set goals or progress criteria
post condition	User receive timely reminders and notifications to aid their memorization process.

Figure 1.1: User modules



2.2 Functional requirements

This section describes the functional requirements of the system expressed in the natural language style. This section is typically organized by feature as a system feature name and specific functional requirements associated with this feature. It is just one possible way to arrange them. Other organizational options include arranging functional requirements by use case, process flow, mode of operation, user class, stimulus, and response depending on what kind of technique has been used to understand functional requirements. Hierarchical combinations of these elements are also possible, such as use cases within user classes.

Module 2.2.1: User Profile and Progress Management

1. **User Registration and Login:**

- **FR1:** The system should allow users to register and log in using an email and password or Gmail account.
- **FR2:** The system should support profile customization, including the ability to set a profile picture.

2. **Progress Tracking:**

- **FR3:** The system should track the user's memorized verses and review history.
- **FR4:** The system should display a personalized dashboard with analytics on performance and progress.

3. **Goal Setting:**

- **FR5:** Users should be able to set memorization goals.
- **FR6:** The system should send reminders based on these goals.

Module 2.2.2: Personalized Learning Path

4. **Learning Style Analysis:**

- **FR7:** The system should analyze users' learning styles (listening or reciting).

5. **Adaptive Repetition:**

- **FR8:** The system should adjust the repetition strategy based on the user's performance.

6. **Customized Memorization Plan:**

- **FR9:** The system should generate a personalized memorization plan for each user.

Module 2.2.3: Ayah Segmentation and Memorization

7. **Automatic Segmentation:**

- **FR10:** The system should automatically segment Ayahs into manageable chunks according to user performance.
- **FR11:** The system should be able to follow all the Quran rules (waqf rules/ Ayah stopping rules).

Module 2.2.4: Interactive Memorization Modes

8. **Memorization Mode:**

- **FR12:** The system should offer a mode where users follow guided steps to memorize verses.

9. **Recitation Mode:**

- **FR13:**Users should be able to recite verses in real time.

10. **Listening Mode:**

- **FR14:**The system should provide a listening mode with playback and repeat functions.
- **FR15:**The system should take a test to modify user performance.

Module 2.2.5: Real-Time Recitation Feedback

11. **Speech Recognition:**

- **FR16:**The system should use speech recognition to detect errors in real-time.

12. **Immediate Feedback:**

- **FR17:**Users should receive feedback on their recitation, with errors highlighted and corrections suggested.

Module 2.2.6: Mutashabihat Detection and Management

13. **Mutashabihat Detection:**

- **FR18:**The system should detect similar verses that could confuse users.

14. **Differentiating similar verses:**

- **FR19:**The system should offer a special practice for differentiating similar verses.

Module 2.2.7: Notifications and Reminders

15. **Notifications:**

- **FR20:**The system should send daily, weekly, and monthly reminders for memorization goals.

16. **Goal Achievement Alerts:**

- **FR21:**Users should receive alerts when they achieve their goals.

2.3 Non-functional requirements

These define the system's operational characteristics, including performance, security, and usability.

Performance

1. **Real-Time Feedback:**

- **NFR1:**The system should provide real-time recitation feedback with minimal latency, ensuring smooth user interaction.

2. **Scalability:**

- **NFR2:**The system should handle multiple users concurrently without degradation in performance.

Usability

3. **User-Friendly Interface:**

- **NFR3:**The interface should be intuitive and easy to navigate for users of all ages.

5. **Accessibility:**

- **NFR4:**The system should be accessible for users with varying levels of technical proficiency.

6. **Cross-Platform Compatibility:**

- **NFR5:**The system should work on mobile (iOS and Android).

Reliability

7. **Data Consistency:**

- **NFR6:**The system should ensure that user data, such as memorization progress and profile information, is always consistent and accurate.

8. **Availability:**

- **NFR7:**The system should ensure availability for users at all times.

Security

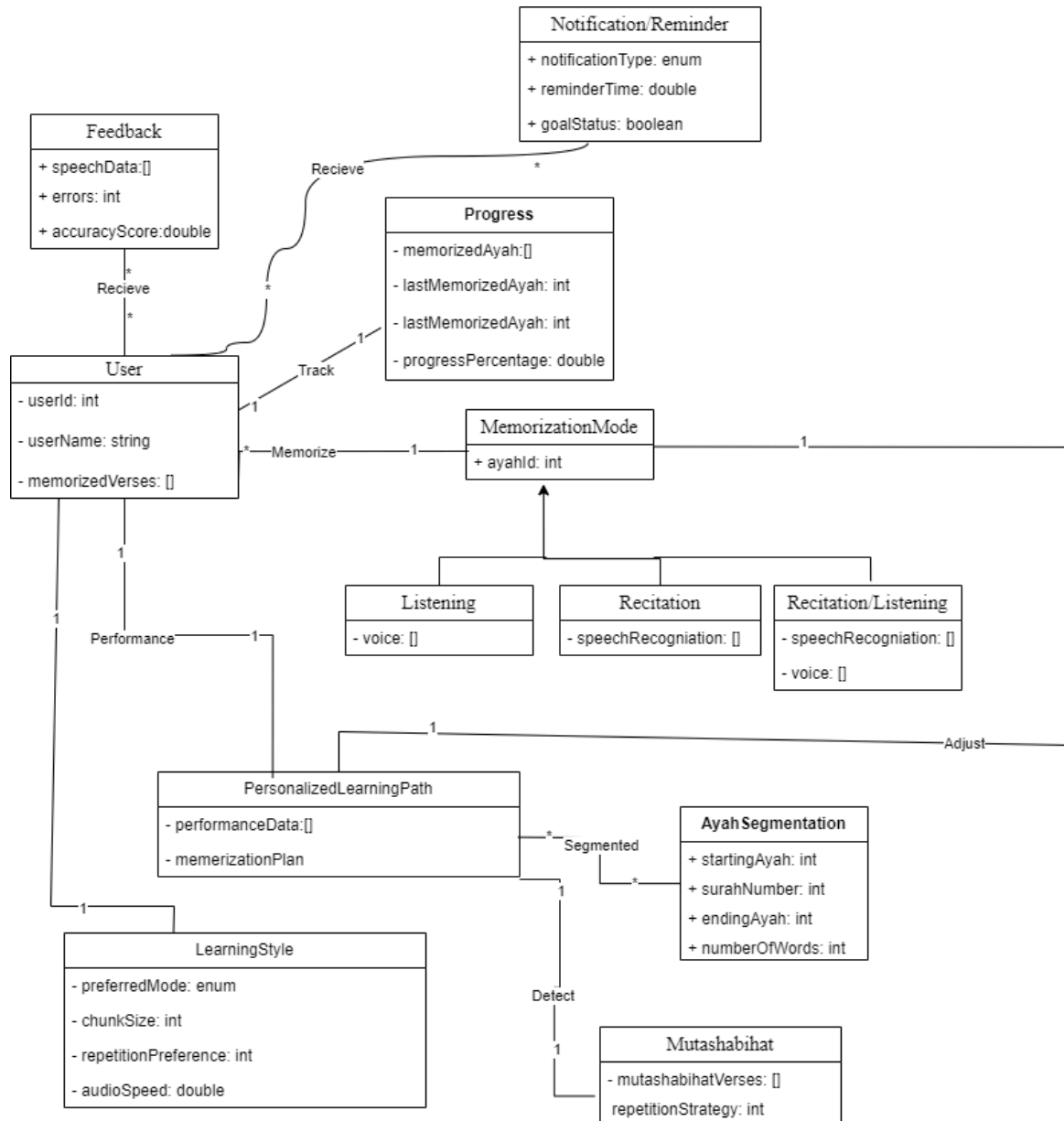
9. **Data Encryption:**

- **NFR8:**User data, including personal information and progress, should be encrypted.

10. **Authentication:**

- **NFR9:**The system should implement secure authentication

2.4 Figure 2.1: Domain model



CHAPTER 3: System overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural design

Develop a modular program structure and explain the relationships between the modules to achieve the complete functionality of the system. This is a high-level overview of how the system's modules collaborate with each other in order to achieve the desired functionality.

Don't go into too much detail about the individual subsystems. The main purpose is to gain a general understanding of how and why the system was decomposed, and how the individual parts work together.

Provide a diagram showing the major subsystems and their connections.

- In initial design stage create Box and Line Diagram for simpler representation of the systems
- After finalizing architecture style/pattern diagram (MVC, Client-Server, Layered, Multi-tiered) create a detailed mapping modules/components to each part of the architecture

3.1.0 Architectural Design for Quran Memorization Assistant System

This section provides a high-level overview of the design of the HifzQuran project. The system is organized into a number of core modules that work together to provide basic functionality that helps users memorize the Quran through personalized learning methods, speech recognition, and real-time Feedback .

3.1.1 Modules Overview

The system is divided into the following primary modules based on the **Model-View-Controller (MVC)** architecture:

- **User Module (Model):** Manages user profiles, preferences, learning styles, and progress. This module is responsible for handling user-specific data such as memorization progress, session history, and personalized settings.
- **Learning Path Module (Model):** Stores and tracks each user's customized learning path based on their progress, performance, and preferences. It dynamically adjusts memorization and repetition cycles for each user.

- **Speech Recognition Module (Model):** Processes and evaluates the user's recitation using speech recognition models. It compares the user's recitation against the Quranic text and provides real-time feedback.
- **Ayah Segmentation Module (Model):** Manages the segmentation of Quranic ayats to support memorization. It splits the verses into manageable parts for repetition based on the user's learning preferences.
- **Similar Ayat Detection Module (Model):** Identifies Quranic verses that have similar phrasing or wording to help users recognize and memorize verses with subtle differences. This module provides warnings and guidance when similar ayat are encountered to minimize confusion during memorization.
- **Feedback Module (Controller):** Gathers data from the Speech Recognition, Learning Path, and Similar Ayat Detection modules and provides performance analysis. It manages interactions between the system's logic and user inputs, ensuring real-time updates for progress and feedback.
- **User Interface Module (View):** Provides a user-friendly interface that enables users to interact with the system, view their progress, access memorization modes, and receive feedback on recitation. It includes various UI components for learning, reviewing, and reciting.

3.1.2 Module Relationships and Interaction

The following diagram (to be provided) illustrates how the major subsystems interact with each other. This high-level overview shows how data flows through the system and how the different modules collaborate to ensure a complete and cohesive Quran memorization process.

- The **User Module** stores individual user preferences and communicates with the **Learning Path Module** to customize learning plans.
- The **Speech Recognition Module** continuously evaluates recitations and sends feedback to the **Feedback**, which processes it and interacts with the **User Interface Module** to display results.

- The **Ayah Segmentation Module** works closely with the **Learning Path Module** to ensure the appropriate verse portions are presented to the user during memorization sessions.
- The **Similar Ayat Detection Module** interacts with both the **Speech Recognition Module** and the **Learning Path Module** to identify and manage verses that are similar, offering real-time notifications to users about potential confusion points.

3.1.3 MVC Architecture

The system follows the **Model-View-Controller (MVC)** architectural pattern due to its modularity and separation of concerns. The reasons for selecting MVC include:

- **Separation of Concerns:** MVC clearly separates the data model (handling user information, learning paths, etc.), user interface (how the user sees and interacts with the system), and controller (which manages the interactions between the model and view).
- **Scalability:** The separation of concerns allows individual modules to be updated or expanded independently. As the system grows (e.g., adding new learning modes, speech recognition features, or similar ayat detection), the existing architecture remains unaffected.
- **Flexibility:** The system's UI, logic, and data handling can be modified or extended without requiring major changes in other areas, ensuring adaptability to future enhancements.

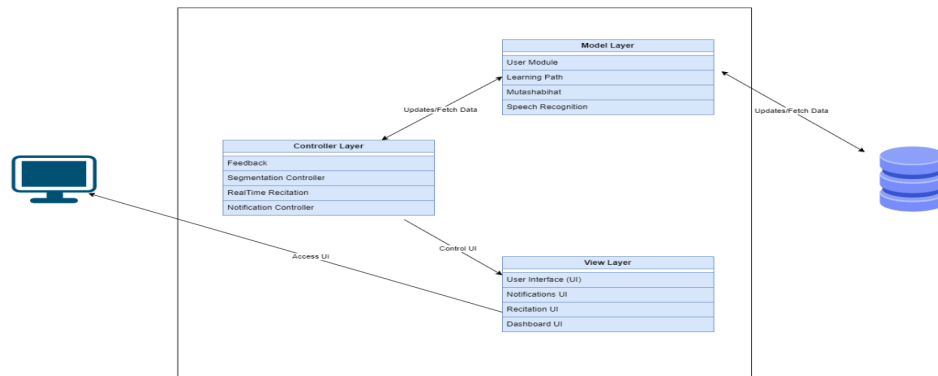
3.1.4 High-Level Box and Line Diagram

Below is a high-level representation of how the system's modules are organized within the MVC architecture.

1. View Layer (User Interface):

- **User Interface (UI):** Displays the personalized dashboard, profile management, and progress tracking.
- **Notifications UI:** Manages user alerts and reminders for memorization progress.

Figure 3.1: MVC



- **Recitation UI:** Provides real-time feedback during recitation, allowing users to track errors and improvements.
- **Dashboard UI:** Displays overall performance analytics, goals, and personalized progress on the dashboard.

2. Controller Layer:

- **Feedback :** This module gathers real-time data and performance metrics, processing it for the View layer to show feedback, such as recitation accuracy or progress.
- **Segmentation Controller:** Manages user interactions regarding ayah segmentation and adjusts the memorization flow.
- **Real-time Recitation Controller:** Manages speech recognition-based interactions, providing feedback on user errors and corrections.
- **Notification Controller:** Handles the delivery of progress-based notifications, reminders, and goal achievement alerts.

3. Model Layer:

- **User Module:** Stores user profile, preferences, and progress data. Responsible for tracking memorized ayahs, sessions, and learning styles.
- **Learning Path Module:** Stores adaptive learning paths, dynamically adjusts memorization strategies based on user performance, and tracks segmented ayah progress.
- **Mutashabihat Module:** Detects and manages similar verses, tracks user progress in overcoming confusing verses, and adjusts memorization strategies accordingly.
- **Speech Recognition Module:** Processes real-time speech input, detects errors in recitation, and records accuracy metrics.

3.1.5 Conclusion

The **MVC architecture** ensures that the HifzQuran application is scalable, and adaptable to future enhancements. Each module is clearly defined and interacts with others through well-defined communication lines. The integration of the **Similar Ayat Detection Module** provides an advanced feature that helps users differentiate between verses that have similar structures, adding to the system's intelligence and making it more efficient for memorizing the Quran. This approach ensures efficient development and maintenance of the system while meeting the functional and operational needs of its users.

3.2 Design models

We will use procedural approach

Figure 3.2: Activity Diagram

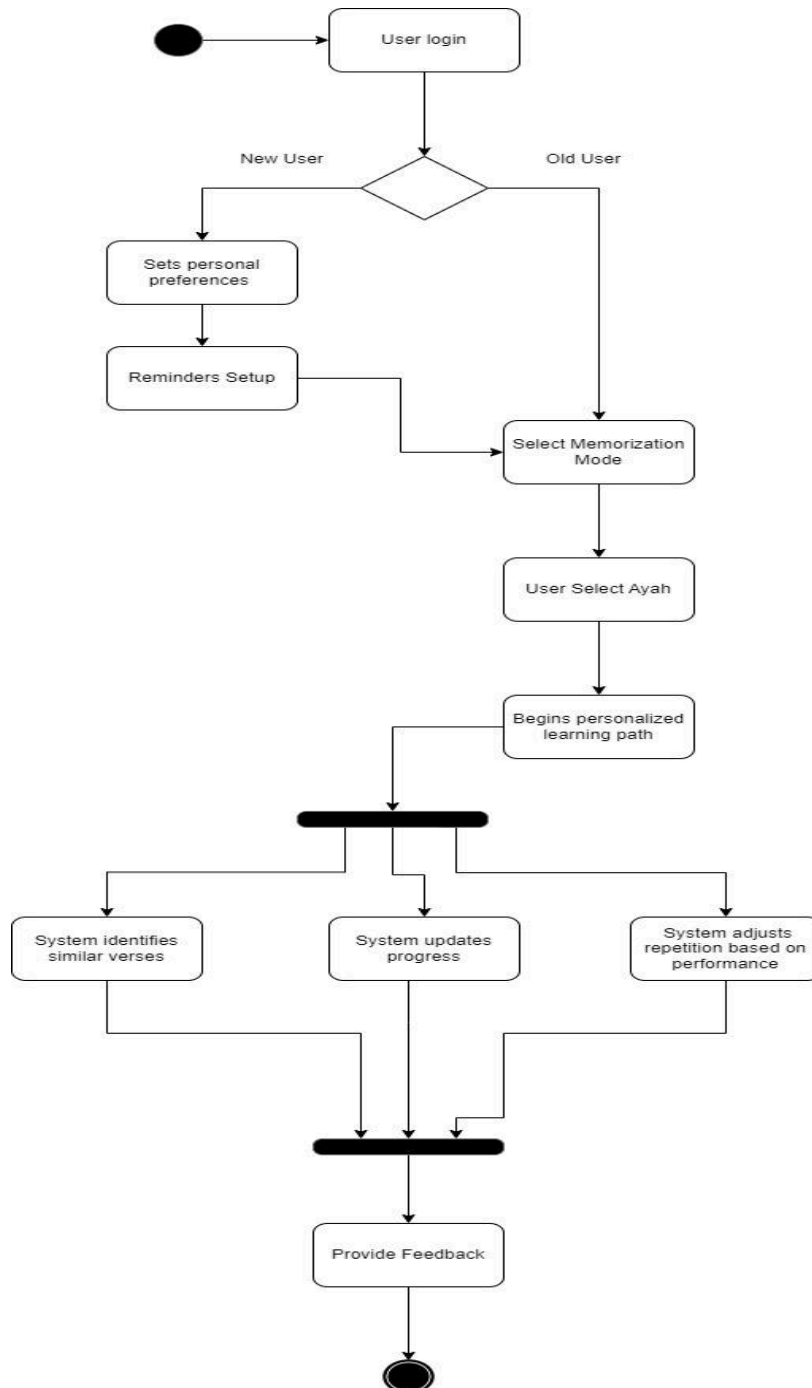


Figure 3.3: Data Flow Diagram

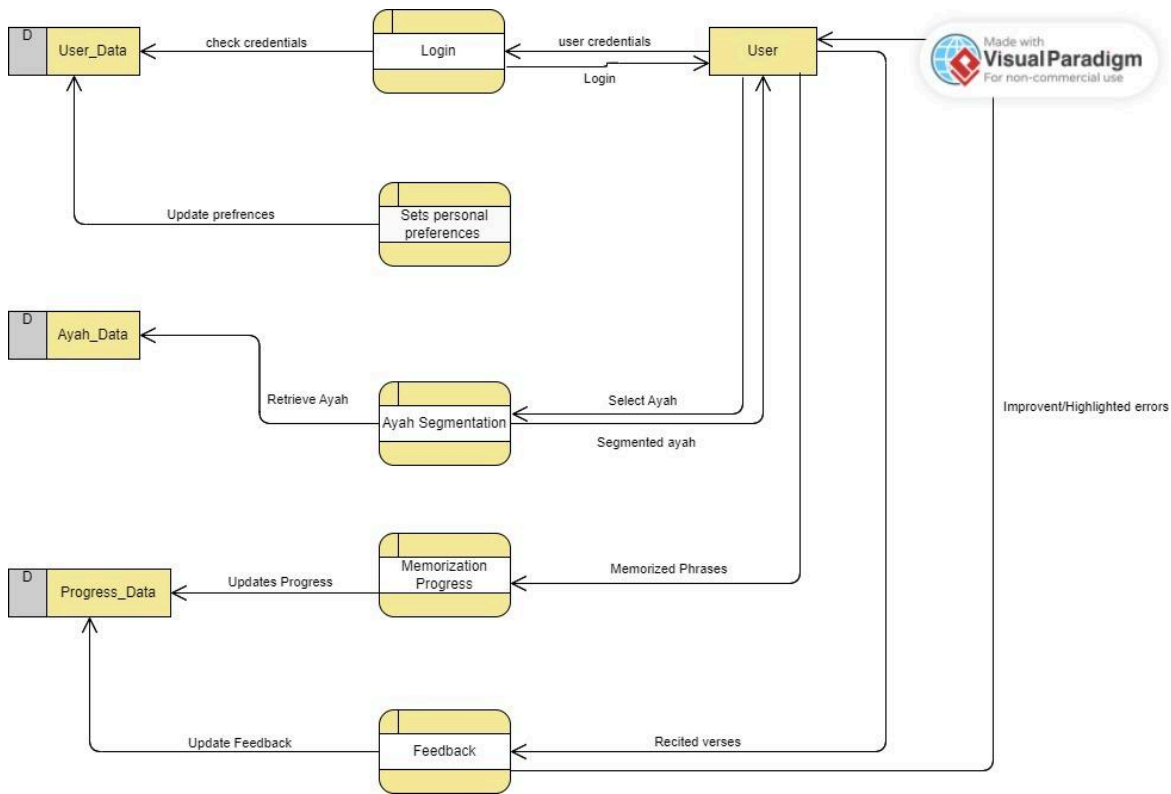
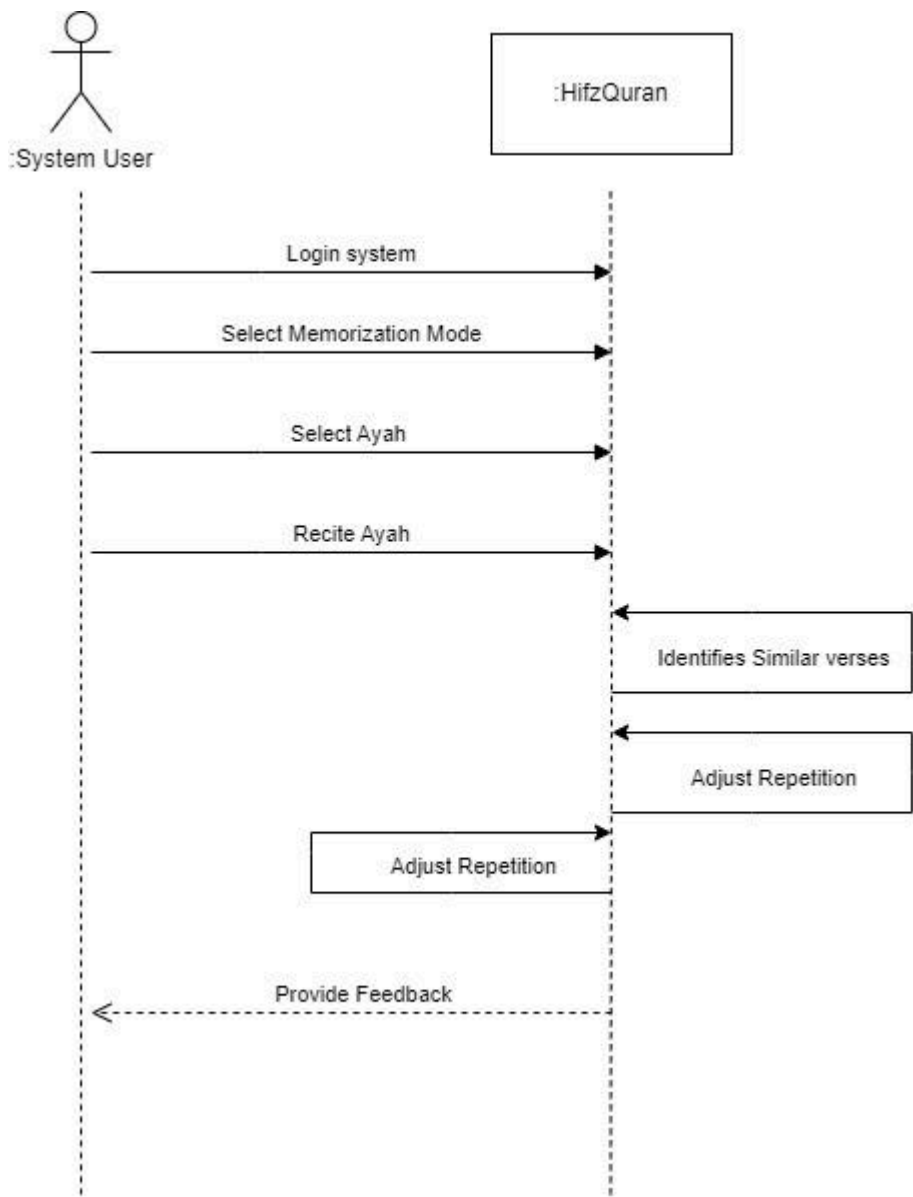


Figure 3.4: System Sequence Diagram



3.3 Data design

In this section, we dive into how the information domain of the Intelligent Quran Memorisation Assistant is transformed into structured data. We describe how the major data entities are stored, processed, and organized within the system. Additionally, we outline the databases and data storage mechanisms utilized to ensure efficient data handling and retrieval.

3.3.1 Introduction

The data design of our application focuses on effectively managing user data, Quranic content, memorisation progress, and system feedback to provide a personalised and adaptive learning experience. The data structures are made to support real-time interactions, adaptive learning paths, and the detection of similar verses (Mutashabihat). Given the sensitive nature of both user information and religious content, data security and integrity are paramount considerations in our design.

3.3.2 Data Storage Technologies

The system employs a hybrid data storage approach, utilizing both relational and NoSQL databases to cater to different data requirements:

- **Relational Database (SQL):** We use a relational database management system (RDBMS) like firebase to store structured data that involves complex relationships and transactions, such as user profiles, progress tracking, and learning paths.
- **NoSQL Database:** For handling unstructured or semi-structured data like speech recognition outputs and session logs, we utilize a NoSQL database like MongoDB, which offers flexibility and scalability.

3.3.3 Data Entities and Structures

The primary data entities in the system and their respective structures are as follows:

- **User Entity**
 - **Attributes:**
 - UserID (Primary Key)
 - Name
 - Email (Unique, Indexed)
 - PasswordHash
 - ProfilePictureURL
 - PreferredLearningStyle (e.g., Listening, Reciting)
 - RegistrationDate

- **Description:** Stores fundamental user information. Passwords are securely stored using hashing algorithms like bcrypt to ensure security.

- **UserProfile**

- **Attributes:**
 - UserID (Foreign Key)
 - Goals (e.g., number of verses per week)
 - Settings (e.g., theme preferences)
 - NotificationPreferences (e.g., reminder timings)
- **Description:** Contains customisable settings and preferences specific to each user.

- **ProgressTracking**

- **Attributes:**
 - ProgressID (Primary Key)
 - UserID (Foreign Key)
 - AyahID (Foreign Key)
 - MemorisationStatus (e.g., Not Started, In Progress, Memorised)
 - LastReviewedDate
 - AccuracyScore (e.g., percentage correct during recitation)
- **Description:** Keeps a record of the user's memorisation progress on each Ayah.

- **LearningPath**

- **Attributes:**
 - PathID (Primary Key)
 - UserID (Foreign Key)
 - CurrentSegment (e.g., AyahID or SegmentID)
 - RepetitionStrategy (e.g., spaced repetition intervals)
 - AdaptiveMetrics (e.g., response times, error rates)
- **Description:** Stores the personalised learning plan for each user, adapting over time based on performance.

- **Ayah**

- **Attributes:**

- AyahID (Primary Key)
 - SurahID (Foreign Key)
 - AyahNumber
 - TextArabic
 - TextTransliteration
 - AudioFileURL
 - SegmentedText (array of text segments)
- **Description:** Contains the content of each Ayah along with metadata for segmentation and playback.
- **Surah**
 - **Attributes:**
 - SurahID (Primary Key)
 - SurahNameArabic
 - SurahNameEnglish
 - NumberOfAyahs
 - **Description:** Represents each chapter of the Quran, providing context and organization for Ayahs.
- **Mutashabihat**
 - **Attributes:**
 - MutashabihatID (Primary Key)
 - AyahID1 (Foreign Key)
 - AyahID2 (Foreign Key)
 - SimilarityScore (e.g., percentage of textual similarity)
 - Notes (guidance on differentiating the verses)
 - **Description:** Stores pairs of similar verses to help users identify and focus on potentially confusing Ayahs.
- **RecitationFeedback**
 - **Attributes:**
 - FeedbackID (Primary Key)
 - UserID (Foreign Key)
 - AyahID (Foreign Key)
 - RecitationAudioURL
 - ErrorsDetected (list of mispronounced words)

- Suggestions (correct pronunciations or tips)
 - Timestamp
- **Description:** Records the results of the speech recognition analysis for each recitation attempt.
- **Notifications**
 - **Attributes:**
 - NotificationID (Primary Key)
 - UserID (Foreign Key)
 - NotificationType (e.g., Reminder, Achievement)
 - Message
 - Timestamp
 - **Description:** Manages all notifications sent to users, whether they are reminders or progress updates.
- **SessionLogs**
 - **Attributes:**
 - SessionID (Primary Key)
 - UserID (Foreign Key)
 - SessionType (e.g., Memorisation, Review)
 - StartTime
 - EndTime
 - ActivitiesPerformed (e.g., Ayahs practiced)
 - **Description:** Logs each user session for analytics and to refine the adaptive learning algorithms.

3.3.4 Data Relationships

The relationships between the entities are crucial for data integrity and efficient retrieval:

- **One-to-One:**
 - **User ↔ UserProfile:** Each user has one profile.
 - **User ↔ LearningPath:** Each user has one personalised learning path.
- **One-to-Many:**
 - **User ↔ ProgressTracking:** A user can track progress on multiple Ayahs.
 - **Surah ↔ Ayah:** Each Surah contains multiple Ayahs.

- **User ↔ RecitationFeedback:** A user can have multiple recitation feedback entries.
- **User ↔ Notifications:** A user can receive multiple notifications.
- **User ↔ SessionLogs:** A user can have multiple session logs.
- **Many-to-Many:**
 - **Ayah ↔ Mutashabihat:** An Ayah can be similar to multiple other Ayahs.

3.3.5 Data Processing and Workflow

The system processes data through various workflows:

- **User Registration and Authentication:**
 - New users register by providing necessary information.
 - Passwords are hashed and stored securely.
 - Upon successful registration, a default UserProfile and LearningPath are created.
- **Profile Customisation:**
 - Users can update their preferences, which are stored in the UserProfile.
 - Changes in preferences may trigger adjustments in the LearningPath.
- **Memorisation Progress Tracking:**
 - When a user practices an Ayah, the ProgressTracking entity is updated.
 - Accuracy scores from RecitationFeedback are used to adjust the memorisation status and repetition intervals.
- **Adaptive Learning Path:**
 - The LearningPath adjusts based on data from ProgressTracking and RecitationFeedback.
 - AdaptiveMetrics are recalculated to personalise the user's learning experience.
- **Ayah Segmentation and Presentation:**
 - Ayahs are segmented based on linguistic rules and stored in the SegmentedText attribute.
 - The system presents these segments during memorisation sessions for easier learning.
- **Mutashabihat Detection and Management:**
 - When a user selects an Ayah, the system checks the Mutashabihat entity for similar verses.

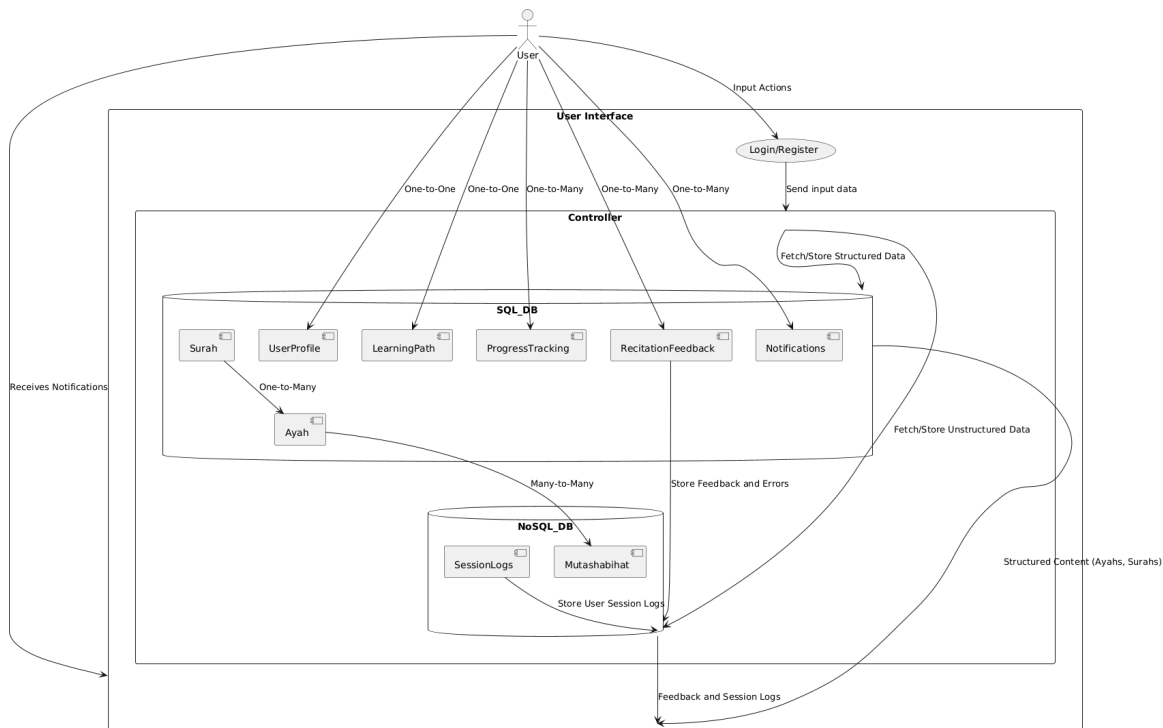
- If similar verses are found, the system provides additional guidance to help differentiate them.
- **Real-Time Recitation Feedback:**
 - The user's recitation is processed by the speech recognition module.
 - Detected errors and suggestions are stored in RecitationFeedback.
 - Immediate feedback is provided to the user via the interface.
- **Notifications and Reminders:**
 - Based on the user's goals and activity, notifications are generated and stored.
 - The system checks the Notifications entity to display unread messages to the user.
- **Session Logging and Analytics:**
 - Each user session is logged in SessionLogs.
 - Data from session logs is used for analytics and to improve the adaptive learning algorithms.

3.3.6 Security and Data Integrity

Ensuring the security and integrity of data is a critical aspect of our design:

- **Data Encryption:**
 - Passwords are hashed using strong algorithms to prevent unauthorized access.
- **Access Control and Authentication:**
 - Role-Based Access Control (RBAC) restricts access to authorized users.
 - Users can only access their own data; administrative functions are protected.
- **Input Validation and Sanitization:**
 - All user inputs are validated to prevent SQL injection and other attacks.
 - Data is sanitized before storage to maintain data integrity.
- **Regular Backups and Recovery:**
 - Automated backups are scheduled regularly.
 - Backup data is encrypted and stored securely to facilitate disaster recovery.
- **Audit Trails:**
 - Changes to critical data are logged.
 - Audit logs help in tracking unauthorized access or modifications.

3.3.7 Figure 3.5 Data Flow Diagram



3.3.8 Conclusion

The data design of our Intelligent Quran Memorisation Assistant is meticulously structured to support a seamless and personalized user experience. By using both relational and NoSQL databases, we accommodate the diverse data types and access patterns required by the system. Robust security measures ensure that user data is protected, while the organized data structures facilitate efficient processing and retrieval. This comprehensive data design underpins the system's ability to provide adaptive learning paths, real-time feedback, and effective memorisation assistance to users

Bibliography

- <https://mabadiiqtishada.org/index.php/EduSpectrum/article/view/27>
- <https://www.almadina.org/studio/articles/using-technology-to-make-quranic-memorization-easier-bilal>
- memon
- <https://quranguest.com/role-of-technology-in-modern-quranic-education/>
- <https://riwaqalquran.com/blog/hafiz-are-in-the-world/>
- [https://en.wikipedia.org/wiki/Hafiz_\(Quran\)](https://en.wikipedia.org/wiki/Hafiz_(Quran))
- <https://firdawsacademy.com/memorizing-the-quran/>
- <https://www.learnreadquran.com/quran-memorization-tips-benefits-and-challenges/>
- <https://howtomemorisethequran.com/new-muslims-guide-memorising-the-quran/>